

RESEARCH ARTICLE

Preventing Defaults in Response to Deteriorating Bank Health: The Prompt Corrective Action Approach

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ABSTRACT

Prior research shows that borrowers are more likely to default when their banks are financially distressed, particularly where contract enforcement is weak. We examine whether regulatory intervention in the form of Prompt Corrective Action (PCA), which seeks to improve bank health through enhanced monitoring, reverses such defaults. To address this question, we exploit the bright-line entry thresholds in India's PCA regime using a regression discontinuity framework. We first show that such defaults exist in India. Our main result is that PCA intervention significantly reduces such defaults. The result is robust to variation in methodology and alternative definitions of bank health. Our evidence suggests that PCA reduces such defaults by credibly signaling to borrowers the likely restoration of bank health and continuity of lending relationships. Its effectiveness was reinforced by an earlier regulatory reform that improved the timeliness of loan-loss provisioning, enhancing the credibility of enforcement. Overall, our findings suggest that PCA, when underpinned by credible financial reporting, can serve as an effective policy tool to curb strategic defaults.

JEL Classification: M41, M48, G21, G28, E58

Prévenir les défaillances en réponse à la détérioration du système bancaire : une approche fondée sur des mesures correctives rapides

RÉSUMÉ

Selon de précédentes études, la probabilité de défaillance des emprunteurs augmente lorsque les banques avec lesquelles ils font affaire connaissent des difficultés financières, notamment lorsque l'exécution des contrats est inefficace. La présente étude examine dans quelle mesure l'intervention réglementaire axée sur des mesures

correctives rapides, conçues pour renforcer le système bancaire par une surveillance accrue, permet de corriger ces défaillances. Pour ce faire, les auteurs définissent les seuils d'entrée du dispositif de mesures correctives rapides instauré en Inde, en s'appuyant sur un modèle de régression sur discontinuité. Ils démontrent, dans un premier temps, l'existence de ces défaillances en Inde. Leur résultat principal indique que l'application des mesures correctives rapides permet de corriger ces défaillances de manière significative. Cette conclusion demeure convaincante face aux variations méthodologiques et aux différentes définitions s'appliquant à l'état du système bancaire. Les données suggèrent que le dispositif de mesures correctives rapides corrige ces défaillances en informant de manière crédible les emprunteurs d'une amélioration attendue du système bancaire et du maintien des relations avec le prêteur. L'efficacité de ce dispositif a été renforcée par une réforme réglementaire antérieure ayant accéléré le processus de provisionnement des pertes sur prêts, augmentant ainsi la probabilité d'exécution. Dans l'ensemble, les résultats suggèrent que les mesures correctives rapides, lorsqu'elles tiennent compte d'informations financières crédibles, peuvent agir comme un mécanisme d'intervention efficace pour prévenir les défaillances d'ordre stratégique.

1 | Introduction

Regulatory enforcement actions by banking supervisors are central to shaping bank behavior and maintaining financial stability.¹ An extensive literature documents that regulatory actions affect various dimensions of bank operations, including lending decisions (Ertan et al. 2017; Balakrishnan et al. 2021; An et al. 2024; Granja and Leuz 2024), financial reporting and disclosure quality (Nicoletti 2018; Costello et al. 2019), and the stability of the banking system (Granja 2018). One such crucial dimension is the repayment behavior of the borrowers. Prior research has shown that borrowers default in response to the deteriorating health of banks (Bond and Rai 2009; Schiantarelli et al. 2020). However, to the best of our knowledge, the literature has not examined whether regulatory interventions targeting bank health—such as Prompt Corrective Action (PCA)—can help mitigate such defaults by borrowers (Jones and King 1995; Delis and Staikouras 2011; Delis et al. 2017; Wheeler 2019; Kleymenova and Tomy 2022; An et al. 2024).² This study fills this gap by analyzing the impact of PCA on borrower repayment behavior within India's regulatory framework.

Our conceptual framework builds on the notion that in environments with weak contract enforcement, lenders rely on the threat of cutting future access to credit to induce repayment (Bond and Rai 2009). When bank health deteriorates, the credibility of this threat declines. Borrowers may then find it attractive to strategically conserve cash (Almeida et al. 2004; Duchin et al. 2010) by defaulting selectively on unhealthy banks (Trautmann and Vlahu 2013). However, defaults also entail costs, including future access to credit from banks and other sources, the threat of insolvency proceedings, and reputational damage from adverse media coverage. These trade-offs are particularly salient in settings with weak contract enforcement, where the expectation of continued lending relationships is an important driver of repayment behavior (Schiantarelli et al. 2020).³ We examine and find that such strategic defaults on weak banks exist in our setting.

Building on this, we next ask our central question: how does PCA shape borrower defaults on banks with deteriorating fundamentals? The effect of PCA on defaults in response to deteriorating bank health is not straightforward. On the one hand, enforcement actions such as PCA may impose lending restrictions on weak banks, reducing the value of continued relationships with them and prompting borrowers to default more (Schiantarelli et al. 2020; Avramidis et al. 2023). On the other hand, PCA may discourage defaults through several channels. First, borrowers may view PCA as a credible intervention aimed at restoring bank health (Aggarwal and Jacques 2001), which increases the value of maintaining the relationship with the bank. Second, PCA banks under regulatory supervision may present a stronger threat of initiating insolvency proceedings (Visaria 2009; Rodano et al. 2016; Iverson 2018). Third, such banks may be more effective in restructuring loans by credibly leveraging the threat of insolvency. Given these competing forces, the net effect of PCA on borrower repayment behavior is ultimately an empirical question. We investigate the above question and find that PCA reduces defaults by borrowers on weak banks.

Our empirical setting uses the PCA framework implemented in India by the Reserve Bank of India (RBI, the country's central bank). In this framework, banks breaching one or more thresholds across five specified accounting and operating

parameters are placed under PCA. These banks face pre-specified and, at times, discretionary regulatory restrictions ranging from curbs on dividends to suspension of lending. The goal is to restore the health of banks subject to the intervention. The Indian setting is ideal for our study for two reasons. First, PCA actions in India are triggered by transparent and rule-based thresholds, unlike settings such as the United States or the United Kingdom, where the admission criteria may be discretionary (Kleymenova and Tomy 2022). The objective cut-offs allow for cleaner identification of the effects of PCA. Second, publicly available, granular bank-firm-level delinquency data allow us to directly examine impacts on borrower repayment behavior.

We start our analysis by examining whether strategic defaults in response to deteriorating bank health exist in India. We use loan performance data from TransUnion CIBIL, the largest credit bureau in India. We construct measures of bank health based on shocks to their borrowers' health. Identifying bank health based on borrowers' health mitigates concerns about reverse causality that arise when loan performance is used both as a proxy for bank health and as the outcome variable. Our evidence confirms the presence of defaults in response to a deterioration in bank health in India.

We then conduct robustness tests to address concerns about the existence of such defaults. First, we address reverse causality concerns by employing a Bartik instrumental variable (IV) approach (Schiantarelli et al. 2020). Second, we verify whether the costs of defaults in our setting are sufficiently low to trigger non-repayment in response to deteriorating bank health. We assess three key costs—insolvency risk, impaired access to future credit, and reputational loss.⁴ We find that they are negligible for a large proportion of firms in our setting, which incentivizes strategic default. Finally, we confirm that our results are not driven by geographical shocks jointly affecting borrower and bank health.

Next, we turn to our main analysis, which investigates whether the PCA policy is effective in curbing defaults on unhealthy banks. The unique institutional design of the PCA framework in India allows us to implement a sharp regression discontinuity (RD) design to test this hypothesis. PCA admissions are triggered by breaches of arbitrary, rule-based accounting thresholds. The banks just above and just below these cut-offs are plausibly similar across both observable and unobservable characteristics. Thus, observed discontinuities at the thresholds help us identify the causal effect of PCA on defaults. We employ the *robust RD* methodology developed by Calonico et al. (2014). The methodology improves inferences by using an objective bandwidth selection procedure and correcting for bias due to bandwidth choice.

For the RD tests, we organize data at the firm-bank-quarter level. The running variable is constructed as the bank's distance from the PCA qualification thresholds in a given quarter.⁵ Since there are five separate accounting metrics, we follow Reardon and Robinson (2012) to create a bank-quarter level *binding* running score. We use the maximum of the standardized values of the five running scores. The treatment (control) group consists of firm-bank-quarter observations for which the bank narrowly qualifies for (misses) the PCA admissions threshold. Thus, within the same firm-quarter, treatment status may vary based on the health of the bank under consideration.

The RD tests show a 3.2 percentage point decline in the delinquency rate on loans from banks that enter the PCA regime. Relative to the unconditional mean delinquency rate of 2.9%, this represents an effect of ~1.1 times the baseline rate. This result suggests that the PCA intervention reduces default in response to deteriorating lender health. Finally, we find that the reduction in defaults is concentrated among firms located in areas with high legal inefficiencies and is unrelated to the health of borrowers.

Finally, we investigate the mechanism underlying the observed reduction in default following PCA. We argue that borrowers' belief in PCA's ability to restore a distressed bank's health—and the resulting increase in the perceived value of a future borrowing relationship—influences strategic default behavior. We provide direct and indirect evidence consistent with this mechanism. We start with direct evidence using the stock market reaction to banks' PCA entry news. If the market anticipates that the banks that enter PCA are expected to recover under RBI supervision, we expect a relatively positive stock market reaction to the news of PCA entry that is reported by more widely read news sources. We find results consistent with this expectation.

We further support our mechanism with several indirect pieces of evidence that are based on the program's design. First, regulatory actions are more effective when banks' financial reporting is accurate and reliable (Bertomeu et al. 2023).

Recognizing the importance of credible accounting information, the RBI conducted an Asset Quality Review (AQR) before implementing the PCA regime (Chopra et al. 2021).⁶ We find that the special audit significantly improved the timeliness of loan loss provisions. This improvement in banks' financial reporting is likely to strengthen borrowers' belief in the effectiveness of PCA intervention, thereby improving their repayment behavior.

Second, if borrowers' belief in the restoration of bank health under PCA is the key mechanism, then the effect should be stronger when PCA is implemented at an earlier stage of financial deterioration. The intuition is that borrower confidence in recovery is higher when the bank is still in the early stages of distress and remains salvageable, compared with when it is already severely impaired. Our findings are consistent with this prediction. Moreover, we verify that bank health improves significantly following PCA admission. This visible improvement reinforces borrowers' ex ante belief in the effectiveness of PCA, thereby reducing their tendency to default on loans from PCA banks.

Third, the PCA likely created alignment between the regulator's objectives and the interests of the government. This alignment, in turn, can mitigate political interference in enforcement actions, thereby supporting the recovery of PCA banks. A major reason regulatory actions can be ineffective is regulators' inability to enforce rules in spirit due to political interference.⁷ However, in the Indian PCA setting, the regulator has the authority to curb lending and restrict branch expansion of PCA banks without yielding to government pressure (Shukla 2018). Such discipline can impose political costs by slowing economic growth in the short term, especially during elections. Our evidence of reduced politically motivated lending by PCA banks is consistent with this mechanism. As a result, PCA incentivizes the government to work toward reviving affected banks, strengthening borrowers' confidence in their recovery.

Fourth, enforcement actions may be ineffective if banks hide their non-performing loans through evergreening (Caballero et al. 2008; Kashyap et al. 2023).⁸ Therefore, enforcement actions that curb evergreening practices can credibly signal a bank's recovery prospects. Under PCA, the RBI conducts strict monitoring that enhances banks' financial reporting transparency (Costello et al. 2019) and curbs suboptimal lending. Consistent with this mechanism, we find a significant decline in evergreening practices under PCA, increasing the credibility of PCA in restoring bank health.

Next, we discuss two alternative explanations. The first is that PCA banks have superior restructuring ability and therefore experience lower defaults than their counterparts. The second is that PCA banks pose a greater threat of referring borrowers to the insolvency process under the Insolvency and Bankruptcy Code (IBC), thereby incentivizing timely loan repayments and reducing defaults. We conduct a battery of robustness tests to demonstrate that these alternative explanations are unlikely to fully account for our results. While our analyses suggest that a significant portion of the reduction in defaults is attributed to a change in borrower expectations, we cannot entirely rule out the possibility that the alternative mechanisms also contribute to the observed decline in defaults.

Our study contributes to several strands of literature. First, we contribute to the literature on banking enforcement actions. Financial regulators, such as the Federal Deposit Insurance Corporation (FDIC) in the United States, issue enforcement actions—known as Enforcement Directions and Orders (EDOs)—against financial institutions that violate regulations or engage in risky practices (Delis and Staikouras 2011; Delis et al. 2017; Kleymenova and Tomy 2022; An et al. 2024). A common enforcement tool used in several jurisdictions is the PCA order, which is triggered by a deterioration in bank health. Prior studies show that such regulatory actions improve risk management, internal governance, lending practices, and credit access for minority borrowers.⁹

The Indian PCA framework provides a unique setting in which enforcement actions are triggered by publicly observable thresholds. This is in contrast to the discretionary and opaque nature of EDOs in the United States. This design enables cleaner identification of the effects of PCA. Our key contribution to the literature on supervisory enforcement using the Indian PCA setting is that we shift the focus from banks to borrowers, examining how enforcement actions influence borrower behavior. While prior studies show that regulatory actions strengthen banks over time, we find that such actions may also incentivize borrowers to reduce defaults.

Our study also speaks to the crucial role of financial reporting in the prudential regulation of banks and the stability of the financial system (Acharya and Ryan 2016; Bischof et al. 2021; Bertomeu et al. 2023). For instance, Bertomeu et al. (2023) document that accounting information and regulatory supervision are complementary. They show that regulatory interventions are more effective when supported by high-quality financial reporting. Conversely, regulator-driven reporting requirements can have adverse effects when implemented ineffectively. This is underscored by studies that document the negative impact of the adoption of the current expected credit loss (CECL) standard in the United

States on lending (Basu et al. 2025; Chen et al. 2025; Granja and Nagel 2025). Specifically, Basu et al. (2025) show that a disconnect between the regulator and the accounting standard setter led to delayed regulatory guidance on CECL. This resulted in a decline in lending and broader economic consequences. Our study reinforces the bright-side view of Bertomeu et al. (2023). We find that the timelier loan-loss provisioning enabled by the AQR enhanced the credibility of accounting information available to the regulator by facilitating timely PCA interventions. These timely actions, in turn, likely strengthen borrowers' confidence in the recovery of bank health and incentivize loan repayment discipline.

Finally, we contribute to the literature on the response of borrowers to deteriorating bank health. Theoretical studies show the existence of an equilibrium in which borrowers default on banks that are expected to fail (Bond and Rai 2009; Carrasco and Salgado 2014). Trautmann and Vlahu (2013) and Schiantarelli et al. (2020) empirically show that borrowers default selectively on banks with weak fundamentals. Complementing this view, studies such as Leary (2009), Becker and Ivashina (2014), and Ivanov and Wang (2024) find that firms actively shift away from distressed banks and sever banking relationships in favor of alternative sources. Together, these findings highlight that borrowers actively respond to perceived bank fragility. Our study shows that a regulatory intervention such as PCA can influence borrower behavior and curb defaults on unhealthy banks.

2 | Institutional Background and Conceptual Framework

2.1 | Institutional Background

PCA is a regulatory enforcement mechanism that imposes restrictions on unhealthy banks to prevent collapses. One of the first major PCA frameworks was implemented in the United States through the *Federal Deposit Insurance Corporation Improvement Act* (FDICIA) of 1991 (Kleymenova and Tomy 2022; An et al. 2024). Its objective was to identify undercapitalized banks with deteriorating financial conditions, address deficiencies by limiting borrowings and growth, and require capital restoration plans. Studies indicate that FDICIA effectively improved bank capital and reduced credit risk (Jones and King 1995). This regulatory tool has since been adopted by regulators worldwide at various times.

A limited version of the PCA framework for banks was first introduced in India in financial year (FY) 2002.¹⁰ Although the framework granted the RBI formal authority to impose restrictions on unhealthy banks, in practice, several institutions with significant regulatory breaches were not placed under PCA.¹¹ Thus, despite having objective criteria, the RBI exercised substantial discretion in invoking the framework.

Further, in response to the Global Financial Crisis (GFC), India introduced regulatory forbearance on provisioning requirements for restructured loans in FY2008. This policy remained in effect until FY2015 (Mannil et al. 2024). The forbearance allowed weak banks to report stronger financials and thereby avoid PCA. Consequently, the older regime is not suitable for examining the impact of the PCA policy.

Recognizing the shortcomings of the earlier framework, the RBI introduced a new PCA regime in FY2018, following the withdrawal of forbearance and completion of the AQR (RBI 2017). The revised regulation sought to bring unhealthy banks under RBI supervision and to implement corrective measures promptly. Crucially, the RBI based the invocation of PCA on objective criteria rather than discretionary judgment. For all practical purposes, the FY2018 framework can therefore be regarded as the effective introduction of the PCA regime in India.

The revised PCA framework introduced explicit, quantifiable criteria for placing banks under regulatory supervision. It established thresholds based on five measures of bank health: the capital adequacy ratio (CRAR), Common Equity Tier I capital ratio (CET1), net non-performing assets ratio (NNPA), leverage ratio (leverage), and return on assets (ROA). These metrics, consistent with Basel III standards, incorporate both on- and off-balance sheet indicators. The precise definitions of these parameters are provided in Panel A of Table 1.

Under the new framework, breaches are classified into three severity levels, each carrying distinct regulatory consequences. Panel B of Table 1 details the annual threshold values for each parameter across the three levels. For example, the minimum CRAR required to avoid a Level II breach was 7.75% in FY2018, 8.375% in FY2019, and 9% in FY2020. A breach of the threshold for any one parameter is sufficient to trigger a bank's entry into PCA.

TABLE 1 | PCA parameters.

Panel A: Definitions of PCA parameters						
Parameter	Definition					
CRAR	The capital adequacy ratio of the bank, calculated as the ratio of total capital to total risk weighted assets (RWA) of the bank in a year					
CET1	The ratio of Common Equity Tier I capital to the total RWA as defined in Basel III guidelines					
NNPA	The ratio of net non-performing assets (non-performing assets adjusted for provisions) over total loans of the bank in a year					
Leverage	The ratio of Tier I capital to the total exposure of the bank as defined in Basel III. Total exposure also comprises off-balance sheet exposures of the bank					
ROA	The ratio of net income to the total assets of the bank in a year					
Panel B: PCA cut-offs						
Year	Threshold level	CRAR	CET1	NNPA	Leverage	ROA
FY2018	I	< 10.25%	< 6.75%	≥ 6%	≤ 4%	< 0 for 2 years
	II	< 7.75%	< 5.125%	≥ 9%	< 3.5%	< 0 for 3 years
	III	< 6.25%	< 3.625%	≥ 12%	< 3.5%	< 0 for 4 years
FY2019	I	< 10.875%	< 7.375%	≥ 6%	≤ 4%	< 0 for 2 years
	II	< 8.375%	< 5.75%	≥ 9%	< 3.5%	< 0 for 3 years
	III	< 6.875	< 4.25%	≥ 12%	< 3.5%	< 0 for 4 years
FY2020	I	< 11.5%	< 8%	≥ 6%	≤ 4%	—
	II	< 9%	< 6.375%	≥ 9%	< 3.5%	—
	III	< 8%	< 4.875%	≥ 12%	< 3.5%	—
Panel C: PCA admissions						
Year	Threshold level	Technical breaches			Banks under PCA	
FY2018	I	16			6	
	II	6			6	
FY2019	I	18			11	
	II	11			11	
FY2020	I	15			5	
	II	5			5	

Note: Panel A defines all of the financial variables used to determine the threshold violations under the PCA policy. Panel B specifies the threshold levels for admission to PCA. Panel C presents the number of PCA violations and admissions by year and threshold level.

A Level I breach results in relatively minor penalties, such as restrictions on dividend distribution and profit remittances. A Level II breach entails more stringent measures, including restrictions on branch expansion, elevated provisioning requirements, and varying degrees of direct and indirect lending curbs. A Level III breach generally triggers all Level II penalties, with the addition of constraints on management compensation and directors' fees.

We do not use Level I breaches for identification for two reasons. First, enforcement of Level I is discretionary.¹² Second, the corrective actions prescribed for Level I breaches are relatively mild and have little impact on lending. In contrast, Level II breaches are strictly enforced. We verify that all banks violating Level II cut-offs were placed under PCA (Panel C of Table 1), and we list these PCA admissions in Table A.1 in the Supporting Information.¹³ Moreover, such violations lead to coercive restrictions. Therefore, Level II is binding and provides a reliable threshold for studying the effects of PCA.¹⁴

2.2 | Conceptual Framework

Our conceptual framework has two components. The first outlines the hypothesis regarding the existence of strategic defaults on unhealthy banks. The second develops the hypothesis on the effect of PCA on such defaults.

2.2.1 | Existence of Defaults on Unhealthy Banks

In environments with inefficient contract enforcement, the threat of discontinuation of lending serves as the primary deterrent to default (Bond and Rai 2009; Schiantarelli et al. 2020). This deterrent loses effectiveness if a bank's ability to extend future credit is impaired due to deteriorating financial health. In such cases, borrowers may choose to default selectively on unhealthy banks. Bond and Rai (2009) refer to this phenomenon as a “borrower run.” An ideal empirical setting to study this behavior is one characterized by slow contract enforcement and high borrower demand for continued credit access. India offers a close approximation to such a setting: it ranks low globally in contract enforcement index and exhibits high demand for credit due to strong GDP growth and existing credit constraints (Banerjee and Duflo 2014).^{15,16}

However, several countervailing economic forces in the Indian context may reduce the incentive for such defaults. First, government-controlled banks (GCBs) hold a substantial share of the banking sector and may benefit from an implicit government guarantee against failure. Consequently, the likelihood of a “borrower run” on these banks may be limited. Second, defaults are reported to credit bureaus, potentially restricting borrowers' future access to credit.¹⁷ Given these offsetting factors, it is necessary to empirically verify whether defaults in response to the deteriorating health of banks exist in India and to identify the drivers of their occurrence. This motivates our first hypothesis, stated below in null form.

Hypothesis 1 (H1). *Borrowers do not default in response to deteriorating health of banks.*

2.2.2 | Effect of PCA on Defaults

Our primary objective is to examine the effect of the PCA framework on the likelihood of such defaults. Since PCA imposes lending restrictions, borrowers may anticipate reduced credit availability from these banks in the short term. This expectation could increase their propensity to default (Bond and Rai 2009; Schiantarelli et al. 2020; Avramidis et al. 2023).

Alternatively, the PCA framework may reduce defaults through three plausible channels. First, since PCA aims to restore the financial health of banks (Aggarwal and Jacques 2001; Acharya 2018), borrowers may revise upward their expectations of the bank's future lending capacity and, consequently, default less. Second, PCA banks, operating under direct central bank supervision, may pose a greater perceived threat of initiating insolvency proceedings against borrowers, thereby deterring defaults (Visaria 2009; Rodano et al. 2016; Iverson 2018). Third, PCA banks may be more effective at restructuring loans as they can credibly threaten borrowers with costly insolvency processes if repayment terms are not met (Kulkarni et al. 2025). This, too, could reduce defaults. Given these opposing forces, the net effect of PCA on borrower repayment behavior remains an empirical question. This motivates our second hypothesis, stated in null form.

Hypothesis 2 (H2). *Borrowers are not less likely to default on banks after their admission to PCA.*

3 | Data and Sample

We obtain quarterly loan-level data from the Ministry of Corporate Affairs (MCA), which records all registered secured loans. The dataset provides the lender and borrower identities, loan amount, loan disbursement date, restructuring date (if any), and final repayment date.

Loan performance data are collected from CIBIL, India's largest credit information company. CIBIL maintains records of all corporate loans exceeding 10 million Indian rupees (INR) where the lending bank has initiated legal recovery proceedings following default. The RBI mandates banks and financial institutions to report such delinquencies to credit

information companies monthly or more frequently. On average, CIBIL's delinquency data account for ~85% of all non-priority sector NPAs reported by banks.¹⁸ Hence, the CIBIL data provide a representative picture of corporate loan delinquencies.

We manually match firm and bank names across the CIBIL and MCA datasets to construct a combined panel of firm–bank pairs and identify delinquent loans. Bank-level accounting information is sourced from the Prowess database maintained by the Centre for Monitoring Indian Economy (CMIE). Data on PCA criteria are obtained from the RBI's official circular (RBI 2017). Information on banks' entry into and exit from PCA is compiled from news articles (see Table A.1).

We also collect data on India's two primary debt recovery mechanisms—the Debt Recovery Tribunals (DRTs) and the IBC. DRTs are fast-track courts that handle asset-based loan recovery cases, typically involving secured loans backed by collateral (Visaria 2009; Vig 2013). Using court-level data from the official DRT website, we construct a measure of court inefficiency based on the pendency rate—the ratio of pending to total cases handled by each DRT. The IBC is a comprehensive, creditor-in-control insolvency resolution framework that allows creditors to initiate insolvency proceedings against defaulting borrowers through National Company Law Tribunal (NCLT) courts (Chakraborty et al. 2022; Hope et al. 2025). We exploit cross-state variation in legal efficiency to assess the likelihood of IBC proceedings against defaulting firms (Chakraborty et al. 2022). District-level data on total and agricultural credit are obtained from the RBI's Database for Indian Economy (DBIE).

Our sample spans four financial years, from FY2018 to FY2021. We exclude financial firms, utilities, and government-owned borrowers. This yields 472,175 firm–bank–quarter observations, covering 17,516 unique firms and 41 unique lenders. Of these, 253,246 observations correspond to non-PCA firm–bank–quarters with available financial information. Across the 580 bank–quarter observations in the sample period, ~18% are PCA bank–quarters. Twelve banks enter PCA during the sample period. The NNPA rate in the sample is 3%. Variable definitions are provided in Table A.2. Sample construction and summary statistics are presented in Panels A and B of Table 2, respectively.

4 | Defaults on Unhealthy Banks

We begin our empirical analysis by testing H1, stated in Section 2.2. Following the approach of Schiantarelli et al. (2020), we estimate an OLS model to examine whether borrowers are more likely to default on loans extended by unhealthy banks.

Schiantarelli et al. (2020) highlight that measures of bank health based on the proportion of a bank's loans in default are vulnerable to reverse causality. Since a bank's loan losses are the aggregation of all borrower defaults, higher borrower defaults can directly worsen a bank's health. Furthermore, for a firm that defaults on one bank but not another within the same quarter, the inclusion of firm $\times$ time fixed effects may not fully address reverse causality. Unobserved, time-varying factors at the firm–bank–year level, unrelated to the bank's health, may also cause a firm to default selectively on certain banks.

We address this concern by constructing a bank health measure that reflects borrowers' underlying health rather than actual default events. Specifically, we classify a borrower as unhealthy if any of its loans were restructured by any lender in the previous quarter. Prior research documents that, in the presence of regulatory forbearance and delayed recapitalization, banks increasingly used restructuring to conceal loan defaults (Chopra et al. 2021; Mannil et al. 2024). In this context, loan restructuring serves as a credible signal of borrower distress during our sample period. Our measure of bank health is the proportion of a bank's outstanding loans to such distressed borrowers, computed at the bank–quarter level.

By measuring bank health based on borrowers' inherent financial condition and including firm $\times$ time fixed effects, we substantially mitigate reverse causality and endogeneity concerns. Deterioration in a borrower's health affects all banks from which it has borrowed, not only the relatively unhealthy ones. Moreover, if a borrower's default were driven solely by its poor health, it should default on all its lenders. The firm $\times$ time fixed effects allow us to compare the same borrower's repayment behavior toward an unhealthy versus a healthy bank in the same period, yielding a cleaner identification strategy.

We test whether loan defaults increase in response to deteriorating bank health by estimating a model where the key explanatory variable is the interaction between a firm's own health and the health of the lending bank. This specification allows us to examine whether the repayment behavior of financially weaker firms is more sensitive to changes in their lenders' health. Our baseline regression equation is as follows:

TABLE 2 | Sample and summary statistics.

Panel A: Sample construction table							
Sample period					FY2018–FY2021		
Number of firms					17,516		
Number of banks					41		
Number of bank-quarter level observations					580		
Number of firm-bank relations					36,954		
Number of firm-bank-quarter observations					472,175		
Number of non-PCA firm-bank-quarters with financial data					253,246		
Number of non-PCA firm-bank-quarters after fixed effects					163,069		
Panel B: Summary statistics							
Bank-quarter summary statistics							
Variable	Obs	Mean	Median	P1	P99	SD	
<i>NPA</i> (in INR billion)	544	245.91	130.98	3.54	1877.65	331.69	
<i>PCAscore</i>	580	0.06	−0.38	−0.96	2.85	6.67	
<i>NNPA</i> (%)	580	3.88	4.68	0.33	16.02	3.67	
<i>CET1</i>	553	11.33	10.75	0.09	27.88	5.95	
<i>ROA</i>	580	−0.41	0.24	−7.63	1.96	6.99	
<i>CRAR</i>	558	13.84	13.38	6.22	29.20	3.86	
<i>Leverage</i>	113	6.72	5.82	3.01	18.9	3.07	
<i>Post_PCA</i>	580	0.22	0	0	1	0.41	
<i>Badfirmshare</i>	580	0.15	0.14	0.08	0	0.40	
Firm-bank-quarter summary statistics							
<i>Outstanding loan</i> (in INR million)		163,069	1700	313	0.6	24,000	8120
<i>New loan</i> (in INR million)		163,069	309	0	0	700	437
<i>Default</i>		163,069	0.03	0	0	1	0.10

Note: Panel A of the table describes the sample used in our tests. It specifies the number of unique firms, banks, bank-quarters, firm-bank pairs, and firm-bank-quarters used in different tests. Panel B provides the summary statistics at bank-quarter and firm-bank-quarter levels. The bank-quarter level variables include the value of the non-performing asset (*NPA* in INR billion); the running variable *PCAscore* as defined in Section 5; *NNPA* as a proportion of outstanding loan (*NNPA*); ratio between Common Equity Tier I capital and risk-weighted assets (*CET1*); *ROA* (*ROA*); capital adequacy ratio (*CRAR*); ratio of Tier I capital to total exposure (*Leverage*); the indicator variable *Post_PCA*, which takes a value of one after a bank enters the PCA regime, and zero otherwise; and *Badfirmshare* as defined in Section 4. The firm-bank-quarter level variables include outstanding loans in INR million, the indicator variable *New loan*, and the indicator variable *Default* as defined in Section 4.

$$Y_{ijt} = \alpha + \beta_1 \text{Badfirmshare}_{jt-1} + \beta_2 \text{Firm_measure}_{it} + \beta_3 \text{Badfirmshare}_{jt-1} \times \text{Firm_measure}_{it} + \beta_4 X_{jt} + \gamma_{it} + \delta_j + \varepsilon_{ijt}, \quad (1)$$

where the dependent variable Y_{ijt} equals one if firm i defaults on bank j in quarter t , and zero otherwise. The sample covers FY2018 to FY2021 and is restricted to non-PCA bank quarters. The variable $\text{Badfirmshare}_{jt-1}$ measures the proportion of bank j 's outstanding loans in quarter $t - 1$ that were extended to unhealthy firms.

The variable *Firm_measure* denotes the borrower-level measure of health. We employ three measures of firm health: (1) *Operating profit margin*—the ratio of EBITDA to the firm's total revenues, (2) *Interest cover ratio*—a measure of short-term solvency, calculated as the ratio of EBIT to interest expense, and (3) *Cash to asset ratio*—a measure of liquidity, calculated as the ratio of cash holdings to total assets.

The vector of bank-level control variables, X_{jt} —total assets, the deposit-to-total-assets ratio, the cash-to-total-assets ratio, and the exposure of bank j to firm i —follows Schiantarelli et al. (2020). The variables γ_{it} and δ_j denote firm $\times$ year-quarter fixed effects and bank fixed effects, respectively. Exploiting the fact that many firms maintain multiple banking relationships, we include firm $\times$ year-quarter fixed effects so that identification comes from within firm-quarter comparisons of banks with high versus low exposure to unhealthy firms.¹⁹

The findings are presented in Columns 1–6 of Table 3. We include the bank-level control variables in the even-numbered columns and cluster the standard errors at the industry level. We first consider the coefficient of *Badfirmshare*. As expected, the coefficient is positive and statistically significant, with a value of 0.13. The standard deviation of *Badfirmshare* is 0.4. Thus, a one-standard-deviation increase in the share of troubled firms in a bank's portfolio is associated with 52 basis points—an economically significant 17%—increase in defaults relative to the unconditional mean. The result rejects H1 and confirms the existence of loan defaults in response to deteriorating bank health. The coefficient of the interaction between *Badfirmshare* and the measure of borrower health is statistically indistinguishable from zero. Thus, the higher loan defaults observed for banks in poor health are unrelated to borrower health.

TABLE 3 | Existence of defaults in response to deteriorating bank health.

Firm measure	<i>Default_{ijt}</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	<i>Operating profit margin</i>		<i>Interest cover ratio</i>		<i>Cash to asset ratio</i>		<i>Court inefficiency</i>	
<i>Badfirmshare_{jt-1}</i>	0.011** (0.005)	0.010** (0.005)	0.011* (0.006)	0.010* (0.006)	0.011** (0.005)	0.010** (0.005)	−0.061*** (0.022)	−0.085*** (0.023)
<i>Badfirmshare_{jt-1}</i> × <i>Operating profit margin_{it}</i>	−0.000 (0.000)	−0.000 (0.000)						
<i>Badfirmshare_{jt-1}</i> × <i>Interest cover ratio_{it}</i>			−0.000 (0.000)	−0.000 (0.000)				
<i>Badfirmshare_{jt-1}</i> × <i>Cash to asset ratio_{it}</i>					−0.000 (0.000)	−0.000 (0.000)		
<i>Badfirmshare_{jt-1}</i> × <i>Court inefficiency_{it}</i>							0.153*** (0.043)	0.192*** (0.044)
Control variables	No	Yes	No	Yes	No	Yes	No	Yes
Firm × Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	163,069	163,069	164,536	164,536	153,798	153,798	216,583	216,583
<i>R</i> ²	0.393	0.393	0.393	0.393	0.393	0.393	0.399	0.399

Note: The table estimates the defaults by borrowers in response to deteriorating bank health. The data are organized at a bank(j)-firm(i)-quarter(t) level for the sample period FY2018 to FY2021 and are limited to year-quarters when the bank is not admitted to PCA. The dependent variable is *Default*, which takes a value of one for the bank-firm-quarters in which the firm defaults on loan repayments to the bank, and zero otherwise. The variable *Badfirmshare* is the one-quarter lagged value of the ratio between the bank's outstanding loan owed by firms that were involved in restructuring and the bank's total outstanding loan. The firm-level measure in Columns 1 and 2 (3 and 4) (5 and 6) is *Operating profit margin* (*Interest cover ratio*) (*Cash to asset ratio*). The firm-level measure in Columns 7 and 8 is *Court inefficiency*, which is calculated as the ratio of the number of pending cases to the overall cases of the DRT court, which has jurisdiction over the firm. We include the control variables—(1) exposure of the bank to the borrower (*Exposure of bank to borrower*), (2) total assets of the bank (*Total asset*), (3) deposit to total asset ratio of the bank (*Deposit to asset ratio*), and (4) cash to total asset ratio of the bank (*Cash to total asset ratio*)—in the even-numbered columns. We also include firm $\times$ quarter fixed effects and bank fixed effects in all columns. The standard errors reported in the parentheses are clustered at the industry level and adjusted for heteroskedasticity. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

We also examine whether loan defaults in response to deteriorating bank health are concentrated in regions with weak contract enforcement. To do so, we replace *Firm_measure* with a measure of court inefficiency (*Court inefficiency*) in Equation (1) and present the results in Columns 7 and 8 of Table 3. We find that higher loan defaults on unhealthy banks are indeed concentrated in areas with legal inefficiencies.

4.1 | Robustness Tests: Existence of Defaults

4.1.1 | Alternative Specifications

Having established that loan defaults increase in response to deteriorating bank health, we conduct a series of robustness tests to address potential endogeneity and methodological concerns. Our results remain robust to (1) using alternative measures of bank health, (2) clustering standard errors at the firm level, (3) ruling out anticipation effects of PCA intervention, (4) addressing identification concerns with a Bartik-style instrument (Schiantarelli et al. 2020), and (5) accounting for common economic shocks and potential co-determination of bank and borrower health. A detailed discussion and supporting empirical evidence for these tests are provided in Section B of the Supporting Information.

4.1.2 | Defaults on Unhealthy GCBs

As stated in Section 2.2, a potential concern is whether the framework of default in response to deteriorating bank health applies to the Indian context, given the significant presence of GCBs. It is important to emphasize that it is not the threat to the bank's survival, but rather its perceived inability to lend due to financial distress, that incentivizes default. We examine this in Section B.6 through two tests. First, we show that deterioration in GCB health is associated with a decline in their lending (Chopra et al. 2021), making them more vulnerable to borrower defaults. Second, we directly test and confirm that the higher defaults on unhealthy banks also hold within the subsample of GCBs.

4.1.3 | Costs of Default for Borrowers

We next assess whether the costs associated with defaults are sufficiently low to incentivize borrowers to default more on distressed banks. Detailed discussions and empirical tests are provided in Section C of the Supporting Information. The first potential cost is the risk of being pushed into the corporate insolvency process under the IBC (Chakraborty et al. 2022; Hope et al. 2025). In Section C, we argue that the requirement of unanimity among multiple stakeholders with conflicting incentives made IBC costly and slow. The average recovery from the bankruptcy proceedings was also low. These facts make IBC plausibly an unattractive enforcement tool, particularly against medium- and small-sized defaulters. Therefore, while the IBC poses some deterrent to strategic default, its overall impact is limited, and defaults on unhealthy banks persist.

A second potential cost is impaired access to future credit due to a decline in the borrower's credit score following delinquency. In our setting, 72% of firms that default on a bank continue to repay their other lenders. Notably, these delinquent firms still have a 5.3% probability of securing a new loan within 1 year from banks they did not default on, compared with an unconditional probability of 26.7% for the full sample. This indicates that while defaults reduce credit access, the penalty is not prohibitive. Moreover, our empirical tests (Table C.1 in Section C) show that borrowers frequently offset the reduced access to bank credit after default by tapping alternative funding through group affiliations (Bertrand et al. 2002; Gopalan et al. 2007) or related-party transactions (Jian and Wong 2010; Kashyap et al. 2023).

Third, defaults can impose reputational costs through adverse media coverage. However, such coverage is typically limited to large, high-profile cases, while defaults by smaller firms rarely draw public attention. Taken together, these observations indicate that, although IBC, reduced credit access, and reputational concerns can all impose meaningful costs on defaulting borrowers, often these costs are outweighed by the benefits of retaining liquidity within the firm (Almeida et al. 2004; Duchin et al. 2010). Thus, defaults can increase in response to a lender's deteriorating health in our setting.

5 | The Reduction in Loan Defaults: RD Analyses

Having shown that loan defaults occur in response to deteriorating bank health, we next test H2 from Section 2.2, which examines whether PCA intervention alters this behavior. Empirically isolating the effect of PCA on defaults presents identification challenges. Direct comparison between defaults on PCA and non-PCA banks is problematic because the two sets of banks may differ systematically. In particular, time-varying bank-level shocks correlated with PCA designation could confound the results. To address this concern, we employ a sharp RD design that compares firm-bank-quarters in which the bank narrowly meets the PCA criteria with those in which the bank narrowly misses them. Given the similarity in their fundamentals, the likelihood of these two groups of banks being differentially affected by time-varying endogenous shocks is low.

Four features of the regulation make it well-suited for an RD design: (1) PCA has well-defined thresholds; (2) all banks that breach PCA thresholds are admitted to PCA within two quarters; (3) the annual AQR audits prior to PCA reduced the scope for manipulation of accounting numbers; and (4) threshold levels vary during the sample period, further randomizing treatment thresholds.²⁰ For example, the Level II threshold for CRAR was 7.75% in FY2018 and 8.375% in FY2019.

To implement the RD design, we create a single running variable from the five PCA parameters, each measured on different scales. A 0.1 increase in CRAR is not directly comparable to a 0.1 increase in leverage. Therefore, following Reardon and Robinson (2012), Becht et al. (2016), and Manchiraju and Rajgopal (2017), we standardize each variable around its cut-off, normalizing the cut-off to zero. The standardized score is the ratio of the extent of the breach to the cut-off value of the parameter. We assign a sign to the standardized score such that a positive score denotes a PCA violation, and a negative score denotes compliance. For example, for NNPA with a cut-off of 9% (for which higher values lead to PCA qualification), an NNPA of 10% receives a score of $(10 - 9) / 9 = 0.11$. For CRAR, with a cut-off of 7.75% (for which lower values trigger PCA), a value of 8.5% receives a score of $(7.75 - 8.5) / 7.75 = -0.1$.

Since PCA is triggered when at least one threshold is violated, we construct a *binding score* based on all five variables (Reardon and Robinson 2012). This binding score (*PCAScore*) is defined as the maximum of the five standardized scores. For instance, a bank with standardized scores of 0.1, 0.2, -0.1, -0.2, and -0.2 for *CRAR*, *CET1*, *NNPA*, *Leverage*, and *ROA*, respectively, violates the first two measures and has a binding score of 0.2.²¹ This approach ensures that all bank-quarters with a breach or a near miss are correctly classified.

We use the *RD robust* methodology developed by Calonico et al. (2014).²² This is the preferred approach for implementing the RD design to estimate the average treatment effect for the following reasons. First, it uses an objective criterion to select a bandwidth around the cut-off. Second, the “rdrobust” estimator corrects for potential bias arising from large bandwidths.

We organize the data at the bank-firm-quarter level. The dependent variable is *Default*, as defined in Equation (1), and the running variable is *PCAScore* as defined above. Note that the treatment variable changes dynamically over time. Therefore, when a PCA bank no longer breaches the cut-off and exits PCA, it moves out of the treatment group; it enters the control group only if it falls within the narrow RD bandwidth after exiting PCA. We present the results in Panel A of Table 4 for the sample period FY2018–FY2021. The three columns report the conventional, bias-corrected, and robust RD coefficients. We focus on Column 1, where the RD coefficient is -3.2%. The likelihood of default above the cut-off is 3.2 percentage points lower than below, suggesting an effect of ~1.1 times the unconditional rate of delinquency. We plot the estimated linear fit in Figure 1 for all firms. We find a significant discontinuity at the cut-off. Thus, we reject H2 and find that firms default *less* on PCA banks.

5.1 | Conventional RD

We also estimate the following *conventional RD* model:

$$Y_{ijt} = \alpha + \beta_1 Treat_{jt} + \beta_2 PCAScore_{jt} + \beta_3 Treat_{jt} \times PCAScore_{jt} + \gamma_{it} + \delta_j + \varepsilon_{ijt}. \quad (2)$$

The dependent variable Y_{ijt} represents *Default* by firm i on bank j in year-quarter t . The running variable is *PCAScore*. The indicator variable *Treat* equals one for bank-quarters that breach the PCA threshold, and zero otherwise. γ_{it} and δ_j

TABLE 4 | Reduction in defaults: Robust RD.

Panel A: Robust RD			
	<i>Default_{ijt}</i>		
	(1)	(2)	(3)
Method	Conventional	Bias-corrected	Robust
RD estimate	−0.032** (0.015)	−0.042*** (0.015)	−0.042*** (0.015)
Observations	19,834	19,834	19,834
Panel B: Conventional RD			
	<i>Default_{ijt}</i>		
	(1)	(2)	
<i>Treat_{jt}</i>	−0.041*** (0.012)	−0.225* (0.121)	
<i>PCAScore_{jt}</i>	0.766*** (0.134)	7.689** (3.240)	
<i>Treat_{jt} × PCAScore_{jt}</i>	−0.541* (0.310)		
Firm × Year-quarter FE	No		Yes
Bank FE	No		Yes
Observations	19,524		1666
<i>R</i> ²	0.003		0.585

Note: This table reports the RD results for the difference in *Default* between PCA and non-PCA bank-quarters. The data are organized at the bank(*j*)-firm(*i*)-quarter(*t*) level for the period FY2018–FY2021 in both panels. In panel A, the coefficients are estimated using the data-driven local polynomial-based robust inference procedure developed in Calonico et al. (2014). We use the default option available for the stata syntax “rdrobust,” which uses the bandwidth selection procedure “mserd,” vce estimator, and “nn3” for estimation. The variable *Default* is an indicator variable set to one if the firm defaults on loan repayments to a bank in a year-quarter, and zero otherwise. In Panel A, Column 1 reports the estimates for the conventional RD approach, Column 2 provides the bias-corrected RD estimates, and Column 3 provides robust bias-corrected RD estimates. In Panel B, we report conventional RD coefficients. The estimates are reported using a 1st-degree polynomial function of the running variable *PCAScore*, which is as defined in Section 5. The variable *Treat* is an indicator variable set to one when *PCAScore* is more than zero, and zero otherwise. We use a bandwidth of 0.051—the bandwidth chosen by the robust RD specification. We include firm × quarter fixed effects and bank fixed effects in Column 2. The standard errors reported in the parentheses are clustered at the industry level and adjusted for heteroskedasticity. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

represent firm × quarter fixed effects and bank fixed effects, respectively. Standard errors are clustered at the industry level. The interaction term captures the slope of the running variable. We present the results for the optimal bandwidth of 0.051, selected using the RD robust methodology, in Panel B of Table 4. We include firm × quarter fixed effects and bank fixed effects in Column 2.

The likelihood of default above the cut-off is 4.1 percentage points lower than below. The results are consistent with Panel A of Table 4.

5.2 | Validating RD Assumptions

Finally, we conduct several tests to verify the RD prerequisites, including the McCrary (2008) test for bunching around the cut-off, comparisons of treated and control banks, and comparisons of firm characteristics on the two sides of the cut-off. Details are provided in Section D of the Supporting Information.

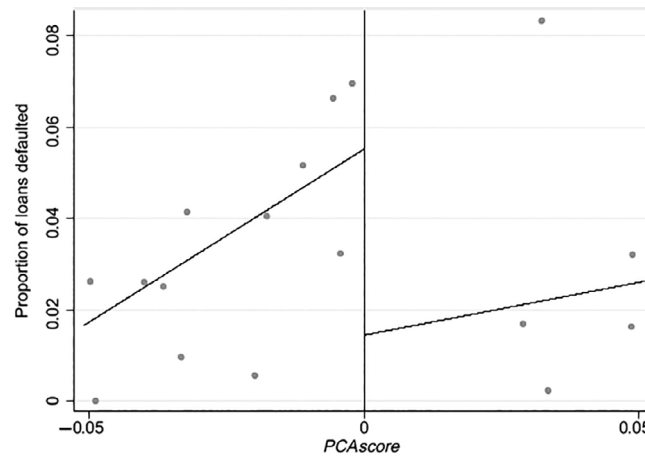


FIGURE 1 | RD plot. The figure shows the RD plot for the difference in default between observations marginally below and marginally above the PCA threshold. The bandwidth is estimated using the procedure developed in Calonico et al. (2014). The data are organized at a bank-firm-quarter level. The x-axis represents the running variable *PCAScore*, which is as defined in Section 5. The y-axis represents the average of *Default* for each bin of observations. The plot uses a first-order local polynomial to present the fitted values on both sides of the PCA cut-off.

5.3 | Cross-Sectional Tests

The estimates in Section 4 indicate that the higher likelihood of default on unhealthy banks is unrelated to borrower health. We therefore expect the reduction in defaults following PCA to exhibit a similar pattern. To test this prediction, we augment the RD specification in Equation (2) with interaction terms between PCA treatment and the measures of firm health:

$$Y_{ijt} = \alpha + \beta_1 Treat_{jt} + \beta_2 Treat_{jt} \times Firm_measure_{it} + \beta_3 Firm_measure_{it} + \beta_4 PCAScore_{jt} + \beta_5 PCAScore_{jt} \times Treat_{jt} + \beta_6 Treat_{jt} \times PCAScore_{jt} \times Firm_measure_{it} + \varepsilon_{ijt}. \quad (3)$$

Here, *Firm_measure* represents the measures of borrower health from Equation (1). The remaining variables are as in Equation (2). The coefficient β_2 captures the differential impact of PCA on delinquencies across borrowers of varying health.

We present the results in Columns 1–3 of Table 5. We find that the coefficient of *Treat* is negative and statistically significant, while the coefficient of the interaction term is statistically indistinguishable from zero. This implies that the reduction in default is unrelated to borrower health. In the spirit of Schiantarelli et al. (2020), these results suggest that PCA intervention mitigates defaults associated with deteriorating lender health.²³ However, it is not possible to conclusively establish the strategic nature of loan defaults.

In Column 4 of Table 5, we examine whether the reduction in defaults is relatively larger in areas with weak contract enforcement. We interact the PCA treatment with the measure of court inefficiency defined in Equation (1). We find that the coefficient of *Treat* becomes insignificant, while that of *Treat* × *Court inefficiency* is negative and significant at the 10% level. As expected, the reduction in defaults is observed in areas with inefficient legal infrastructure, providing further evidence consistent with reversals in strategic defaults due to PCA.

5.4 | Robustness Tests: Reduction of Defaults

5.4.1 | Robustness of RD Estimates: Alternative Specifications

Building on the baseline RD results that document a decline in defaults following PCA, we examine the stability of these estimates using alternative specifications to address potential measurement and identification concerns. The results remain qualitatively similar to the baseline when we use an alternative binding score that incorporates the variability of the accounting variables used in PCA admission, when we cluster the standard errors at the firm level, and when we implement a doughnut hole RD test to address potential manipulation near the cut-off. Supporting evidence is provided in Section E of the Supporting Information.

TABLE 5 | Reduction in defaults and firm heterogeneity.

Firm health measure	<i>Default_{ijt}</i>			
	(1) <i>Operating profit margin</i>	(2) <i>Interest cover ratio</i>	(3) <i>Cash to asset ratio</i>	(4) <i>Court inefficiency</i>
<i>Treat_{jt}</i>	−0.027*** (0.010)	−0.032*** (0.011)	−0.025** (0.010)	0.100** (0.040)
<i>Treat_{jt} × Firm_measure_{it}</i>	−0.026 (0.020)	0.000** (0.000)	−0.000 (0.000)	−0.240*** (0.068)
<i>PCAscore_{jt}</i>	0.430*** (0.100)	0.528*** (0.120)	0.405*** (0.120)	0.188 (0.413)
<i>Treat_{jt} × PCAscore_{jt}</i>	−0.178 (0.272)	−0.322 (0.296)	−0.271 (0.261)	−2.166* (1.278)
<i>Firm_measure_{it}</i>	0.008* (0.005)	−0.000** (0.000)	−0.000 (0.000)	0.061 (0.038)
<i>PCAscore_{jt} × Firm_measure_{it}</i>	0.242* (0.138)	−0.002 (0.002)	0.001 (0.003)	1.424 (1.078)
<i>Treat_{jt} × PCAscore_{jt} × Firm_measure_{it}</i>	0.105 (0.456)	0.001 (0.002)	0.001 (0.008)	1.805 (2.483)
Observations	14,594	14,063	13,369	19,494
<i>R</i> ²	0.010	0.006	0.002	0.004

Note: The table shows the estimates of the effect of firm heterogeneity on the decline in defaults due to PCA. The data are organized at a bank-firm-quarter level for the period FY2018–FY2021. The dependent variable *Default* takes a value of one for the bank-firm-quarters in which the firm defaults on loan repayments to the bank, and zero otherwise. The firm measures in Columns 1, 2, and 3 are *Operating profit margin*, *Interest cover ratio*, and *Cash to asset ratio*, respectively. The firm measure in Column 4 is *Court inefficiency*, calculated as the ratio of the number of pending cases to the overall cases of the DRT court, which has jurisdiction over the firm. The estimates are reported using an RD design with 1st-degree polynomial function of the running variable *PCAscore*. The variable *PCAscore* is as defined in Section 5. The variable *Treat* is an indicator variable set to one when *PCAscore* is more than zero, and zero otherwise. We use a bandwidth of 0.051 across all the columns. The standard errors reported in parentheses are clustered at the industry level and adjusted for heteroskedasticity. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

5.4.2 | Heterogeneity in Borrowers of GCBs and Non-GCBs

One concern with our findings is that borrower profiles may differ systematically between GCBs and non-GCBs. Since most banks breaching the PCA cut-off are GCBs, it is possible that differences in borrower characteristics drive our results.

We conduct two robustness tests to address this concern. First, we reestimate the RD specification after restricting the sample to GCBs. As reported in Table A.4, the results remain qualitatively unchanged. Second, we implement an entropy balancing technique to match borrowers of GCBs and non-GCBs on observable characteristics, and we obtain results that are similar to the baseline (Table E.4 in Section E.4).

5.4.3 | Alternative Identification Approach: Difference-in-Differences

Since the banks entered PCA in a staggered manner, the setting lends itself well to a staggered difference-in-differences (DiD) design. We test the robustness of the RD results by implementing a staggered DiD approach using the following specification:

$$Y_{ijt} = \alpha + \beta_1 \text{Post_PCA}_{jt} + \beta_2 X_{jt} + \gamma_{it} + \varepsilon_{ijt}. \quad (4)$$

Here, $Post_PCA_{jt}$ equals one in periods after bank j is placed under PCA, and zero otherwise. X_{jt} denotes the vector of bank-year-level control variables listed in Equation (1), and the remaining variables are as in Equation (2). The coefficient β_1 captures the differential likelihood that the same borrower defaults on PCA banks relative to non-PCA banks.

Results reported in Table A.5 indicate that the coefficient of $Post_PCA$ is negative and statistically significant. Consistent with the RD estimates, these findings suggest that firms are less likely to default on PCA banks.

6 | Underlying Mechanism

Next, we examine the mechanism underlying the reversal of defaults in response to deteriorating bank health. We present both direct and indirect evidence consistent with the explanation that borrowers expect banks' health to be restored under PCA.

6.1 | Media Coverage and Market Reaction

Our direct evidence for the proposed mechanism is based on the market reaction to news of banks' PCA entry. If the decline in defaults on PCA banks is driven by borrowers' expectations that banks will recover under PCA, we would expect the market to react positively to news of PCA entry. This positive reaction should be more pronounced for PCA announcements appearing in popular media outlets, as borrowers are more likely to observe these and adjust their repayment behavior accordingly.

Therefore, we examine how market reactions differ depending on the news source of PCA disclosure. Among the 12 PCA banks, we find that for eight of them, news of their PCA admission was reported immediately in *The Economic Times*, India's largest business news outlet. Using this bifurcation, we create two market-weighted portfolios: a "high-information access bank portfolio" comprising the PCA banks covered by *The Economic Times*, and a "low-information access bank portfolio" comprising the remaining four PCA banks.²⁴ We then compute the announcement-day returns of both portfolios on the day of PCA entry. We find that the return is higher by 293 basis points for the "high-information access portfolio." This result suggests that markets react more positively to PCA entry news for banks receiving coverage in widely read news outlets.²⁵

6.2 | Indirect Evidence

We next provide indirect evidence based on the design of the PCA policy.

6.2.1 | Timely Intervention

The success of PCA, which aims to rehabilitate unhealthy banks, depends on the timely identification of bank distress. If banks can conceal asset-quality problems through misclassification and under-provisioning, they may evade or delay regulatory intervention, thereby undermining the policy's effectiveness. This makes the quality of financial reporting critical to the success of regulatory supervision (Bischof et al. 2021; Bertomeu et al. 2023).

Recognizing this, the regulator introduced the AQR, a regulator-led special audit, prior to implementing the PCA framework (Chopra et al. 2021). The AQR required banks engaging in discretionary loan-loss provisioning to revise their reported provisions to reflect true asset quality, thereby limiting their ability to delay loss recognition. Notably, annual AQRs conducted from FY2016 to FY2019 resulted in upward revisions of NPAs and provisions by 52% and 30%, respectively. Building on this, we formally test whether the AQR improved the timeliness of loan-loss reporting, following Bushman and Williams (2012). We estimate the following specification using data spanning FY2013 to FY2021.²⁶

$$LLP_{jt} = \alpha + \beta_1 NPA_{jt+1} \times Post_t + \beta_2 NPA_{jt+1} + \beta_3 Post_t + \beta_4 X + \gamma_j + \delta_t + \epsilon_{jt}. \quad (5)$$

Here, the variable LLP_{jt} represents the ratio of provisioning to total advances for bank j in year t . The variable $Post$ equals one for years after FY2016, when the AQR was conducted, and zero otherwise. The vector X contains the set of control variables in Bushman and Williams (2012): NPA in that year, NPA in the previous year, total assets (*Total asset*), capital adequacy ratio (*CRAR*), and profit before provisioning and taxes. The coefficient of interest, β_1 , captures the differential increase in the timeliness of provisioning for future NPAs following the AQR.

We present the results in Table 6. We find that the coefficient of $NPA_{jt+1} \times Post$ is positive and statistically significant, indicating that the AQR enhanced the timeliness of loan-loss provisions. An extensive literature documents that financial reporting quality improves the effectiveness of regulatory interventions (Bertomeu et al. 2023). Hence, the improved timeliness in loan-loss provisions can signal to borrowers a stronger likelihood of bank recovery under PCA, thereby reducing default rates.

Moreover, early detection and action are critical, as banks admitted too late into PCA may be beyond recovery. The RBI's decision to conduct the AQR prior to PCA increases the likelihood of timely admission of weak banks, thereby strengthening borrowers' confidence in PCA's ability to restore bank health. Consistent with this prediction, we test whether the reduction in defaults following PCA varies with the severity of deterioration in bank health at the time of intervention.

6.2.2 | Evidence

To investigate this hypothesis, we employ the measure of the extent of PCA breach using $PCAscore$ at the time of PCA breach. If timely admission matters for borrowers' repayment behavior, we would expect the decline in defaults to be driven by banks that entered PCA with a marginal breach. To test this hypothesis, we estimate the following specification using a sample restricted to banks that entered PCA:

TABLE 6 | AQR and loan-loss provisioning.

	LLP_{jt}	
	(1)	(2)
$Post_t \times NPA_{jt+1}$	0.133** (0.051)	0.116** (0.048)
NPA_{jt+1}	-0.223*** (0.045)	-0.212*** (0.041)
NPA_{jt}	0.472*** (0.044)	0.442*** (0.036)
NPA_{jt-1}	0.254*** (0.092)	0.251*** (0.091)
Control variables	No	Yes
Bank FE	Yes	Yes
Year FE	Yes	Yes
Observations	345	345
R^2	0.837	0.846

Note: This table shows the estimates for tests of the change in timeliness of loan-loss provisioning post-AQR. The data are organized at the bank-year level for the period 2013–2021. The dependent variable, LLP_{jt} , represents the ratio between provisioning and total advances for bank j and year t . The variable NPA_{jt+1} represents the non-performing loans ratio of bank j in year $t + 1$. The indicator variable $Post$ takes a value of one after 2016 or the start of AQR, and zero otherwise. We include the variables representing NPA in the same year (NPA_{jt}) and NPA in the previous year (NPA_{jt-1}) in both columns. We include the control variables—total assets (*Total asset*), capital adequacy ratio (*CRAR*), and profit before provisioning and taxes—in Column 2. The standard errors reported in parentheses are clustered at the bank level and adjusted for heteroskedasticity. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

$$Y_{ijt} = \alpha + \beta_1 \text{Marginal_breach}_j + \beta_2 \text{Egregious_breach}_j + \beta_3 X_{jt} + \gamma_{it} + \varepsilon_{ijt}. \quad (6)$$

The variable *Marginal_breach* (*Egregious_breach*) takes the value of one for the post-PCA period if the bank has a lower (higher) than the median value of *PCAScore* during PCA entry, and zero otherwise. The other details are as in Equation (1). As per our hypothesis, we expect a negative coefficient on *Marginal_breach*, and the coefficient of *Egregious_breach* to be insignificant.

The estimates presented in Table 7 are consistent with our arguments. We find that the coefficient of *Marginal_breach* is negative and statistically significant, while the coefficient of *Egregious_breach* is statistically indistinguishable from zero. Overall, the results indicate that PCA is effective in reducing loan defaults when the intervention occurs at relatively earlier stages of deterioration in bank health.

6.2.3 | Alignment Between the Regulator's and the Government's Interests

Political interference often undermines regulatory effectiveness and the enforcement of prudential rules. For instance, Correia (2014) and Knechel and Park (2022) show that political connections weaken oversight by the SEC and PCAOB, respectively. Similarly, other studies document how political ties distort lending, impede banking resolution, and weaken supervisory discipline (Sapienza 2004; Duchin and Sosyura 2012; Koetter and Popov 2021; Huang and Thakor 2024). In the Indian context, regulatory initiatives to expand credit bureau coverage and strengthen credit discipline have frequently clashed with politically motivated debt waiver programs, reducing their likelihood of success (Kanz 2016; Giné and Kanz 2018; Mukherjee et al. 2018; Tantri 2018).

However, PCA regulation stands out because it grants full authority to the RBI to control banks' lending and other operations. Under PCA, the RBI can direct banks to completely halt lending to some or all sectors. This power allows the RBI to counteract the adverse effects of political interventions by curbing politically motivated lending (Shukla 2018).

Furthermore, the RBI may sometimes prioritize objectives such as price or financial stability over short-term economic growth, necessitating temporary lending restrictions under PCA. A reduction in lending, even if temporary, can impose political costs, making it difficult for political leaders to convince the electorate and attribute economic distress to regulatory actions. Given the powers conferred to the RBI under PCA, it is also challenging for the executive to nullify its effects.

TABLE 7 | Healthy versus unhealthy PCA banks.

	<i>Default_{ijt}</i>	
	(1)	(2)
<i>Marginal_breach_j</i>	−0.023*** (0.008)	−0.018** (0.007)
<i>Egregious_breach_j</i>	−0.011 (0.016)	0.003 (0.015)
Control variables	No	Yes
Firm × Year-quarter FE	Yes	Yes
Observations	44,027	44,027
<i>R</i> ²	0.405	0.406

Note: The table compares the reversal in defaults during the PCA regime across relatively healthy and unhealthy PCA banks. The data are at the firm-bank-quarter level, and we restrict the sample to PCA banks only. The dependent variable *Default* takes a value of one when the firm defaults on the bank, and zero otherwise. The variable *Marginal_breach* takes a value of one for the post-PCA period if the bank has a lower than the median value of *PCAScore* at the time of entry into PCA, and it takes a value of zero in the pre-PCA period. The variable *Egregious_breach* takes a value of one for the post-PCA period if the bank has a higher than the median value of *PCAScore* at the time of entry into PCA, and it takes a value of zero in the pre-PCA period. We include bank-year level control variables listed in Table 3 in the even-numbered column and firm × quarter fixed effects in all columns. The standard errors clustered at the industry level and adjusted for heteroskedasticity are reported in parentheses. ***, **, and * represent statistical significance at 1%, 5%, and 10%, respectively.

Thus, it is in the ruling government's interest to cooperate with regulators to restore banks' health and exit PCA. This alignment of incentives between regulators and the government is a unique feature of the PCA regulation, which can help reduce defaults by altering borrower expectations regarding the eventual recovery of banks.

We provide evidence consistent with the above mechanism—that PCA enforcement reduces lending, including in politically important segments—thereby aligning the interests of the regulator and the government. This, in turn, can potentially increase the likelihood of restoring bank health.

First, we examine whether overall credit by banks declines after they are placed under PCA restrictions. We use the following specification for this test:

$$Y_{ijt} = \alpha + \beta_1 \text{Post_PCA}_{jt} + \beta_2 X_{jt} + \gamma_{it} + \delta_j + \varepsilon_{ijt}, \quad (7)$$

where Y_{ijt} is the natural logarithm of loans extended by bank j to firm i . The indicator variable Post_PCA equals one after a bank enters the PCA regime, and zero otherwise. We include control variables, X_{jt} , defined in Equation (1). The results are presented in Panel A of Table 8. All columns include control variables, firm $\times$ quarter fixed effects, and bank fixed effects. We find a negative association between Post_PCA and lending, indicating that banks placed under PCA restrictions reduce lending.

Second, we examine whether the decline in lending by PCA banks leads to a reduction in credit in the regions where they dominate. We estimate the following equation:

$$Y_{kt} = \alpha + \beta_1 \text{DistrictPCAexposure}_{kt-1} + \gamma_k + \delta_t + \varepsilon_{kt}. \quad (8)$$

Here, the data are at a district-year level. The dependent variable Y_{kt} is the natural logarithm of outstanding credit in district k in year t . $\text{DistrictPCAexposure}$ is an indicator variable equal to one when the district lies above the median in terms of the proportion of credit from PCA banks in year $t - 1$, and zero otherwise. The data spans FY2016 to FY2021.²⁷ Results are presented in Column 1 of Panel B of Table 8. As expected, the coefficient of $\text{DistrictPCAexposure}$ in Column 1 is negative and significant. Lending to districts with high exposure to PCA banks declines by 5.3 percentage points.

Third, we examine whether agricultural credit also declines. Nearly half of India's population depends on agriculture, making agricultural credit politically sensitive. We show in Column 3 of Panel B of Table 8 that agricultural lending to districts with high exposure to PCA banks also declines by 4.5 percentage points.

Finally, we examine whether the decline in credit due to PCA reverses during elections. Cole (2009) shows that bank lending close to elections is politically more significant. Therefore, if PCA banks continue to lend less even during elections, the political costs of reduced lending may escalate. To test this, we modify Equation (8) by adding the interaction between $\text{DistrictPCAexposure}$ and an indicator variable Election , which equals one if the district's state has a scheduled election during the year, and zero otherwise.

We present the results in Columns 2 and 4 of Panel B of Table 8. The coefficient of $\text{DistrictPCAexposure}$ is negative and statistically significant, while the coefficient of the interaction is insignificant. Thus, the ruling government seems unable to pressure PCA banks to increase lending during elections.²⁸ Consequently, the government has an incentive to work with the regulator to improve bank health and facilitate banks' exit from PCA. We also contribute to the literature that documents distortions created due to political interests (Sapienza 2004; Correia 2014) by showing that the PCA enforcement is critical to the efficacy of the intervention in restoring the health of distressed banks.

The above results support the mechanism that PCA acts as a credible signal of the restoration of banks' health. Moreover, consistent with the ex ante expectation that such alignment should facilitate bank recovery, we observe ex post improvements in bank health following PCA. As shown in Figure A1 and Panel A of Table A.6, PCA banks experience a 4.3 percentage point increase in the capital adequacy ratio along with significant gains in other parameters. Furthermore, Panel B of Table A.6 shows that banks exiting PCA are more likely than non-PCA banks to expand lending, underscoring the policy's role in restoring their operational capacity.

TABLE 8 | Decrease in lending by PCA banks.

Panel A: Firm-level lending				
	<i>Log loan_{ijt}</i>		<i>New loan_{ijt}</i>	
	(1)		(2)	
<i>Post_PCA_{jt}</i>	−0.165*** (0.045)		−0.009*** (0.002)	
Firm × Year-quarter FE	Yes		Yes	
Bank FE	Yes		Yes	
Observations	310,490		310,490	
<i>R</i> ²	0.379		0.386	
Panel B: District-level lending				
	<i>Log Total credit_{dt}</i>		<i>Log Agriculture credit_{dt}</i>	
	(1)	(2)	(3)	(4)
<i>Post_PCA_{jt} × DistrictPCAexposure_{dt}</i>	−0.053*** (0.016)	−0.051*** (0.016)	−0.045** (0.021)	−0.038* (0.021)
<i>Election_{st}</i>		0.022* (0.011)		0.041*** (0.015)
<i>Post_PCA_{jt} × DistrictPCAexposure_{dt} × Election_{st}</i>		−0.016 (0.020)		−0.044 (0.032)
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	2245	2245	2245	2245
<i>R</i> ²	0.973	0.973	0.950	0.950

Note: This table shows the impact of PCA regulation on lending activities. In Panel A (B), the data are organized at a bank-firm-quarter (district-year) level and span the years FY2018 (FY2016) to FY2021. The dependent variable in Column 1 of Panel A is *Log loan*, which is the natural logarithm of the amount of the loan borrowed by the firm from the bank in the respective year-quarter. The dependent variable in Column 2 of Panel A is *New loan*, an indicator variable that takes a value of one if the bank extends a loan to the firm in the year-quarter, and zero otherwise. The explanatory variable *Post_PCA* takes a value of one after a bank enters the PCA regime, and zero otherwise. We include control variables listed in Table 3, firm × quarter fixed effects, and bank fixed effects in all columns of Panel A. The dependent variable in Columns 1 and 2 (3 and 4) of Panel B is the natural logarithm of *Total credit* (natural logarithm of *Agriculture credit*) in the district. The indicator variable *DistrictPCAexposure* takes a value of one if the district lies above the median in terms of the proportion of its total outstanding loan owed to PCA banks in the previous year, and zero otherwise. The indicator variable *Election* takes a value of one if the state in which the district is located has an election in the year, and zero otherwise. We include district and year fixed effects in all columns of Panel B. The standard errors reported in the parentheses are clustered at the industry (district) level and adjusted for heteroskedasticity in Panel A (B). ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

6.2.4 | Potential to Curb Loan Evergreening

Indiscriminate loan evergreening, facilitated by regulatory forbearance, was a key driver of the banking crisis in India (Chari et al. 2021). Kashyap et al. (2023) document its persistence even after the forbearance was withdrawn. Prior research shows that evergreening increases the likelihood of bank failure (Peek and Rosengren 2005; Caballero et al. 2008), which can, in turn, fuel borrower expectations of bank failure and incentivize defaults. Preventing such practices is therefore critical to reversing these defaults.

While most ex post regulatory tools lack sufficient oversight of day-to-day lending, the PCA framework empowers the regulator to restrict suboptimal lending by PCA banks. Borrowers may thus view PCA as a credible tool to halt the deterioration of bank health (evergreening) and adjust repayment behavior accordingly. Consistent with this interpretation, evidence in Section F.1 in the Supporting Information indicates that PCA intervention addresses the root causes of the non-performing loans crisis by (1) reducing evergreening and (2) curtailing diversion of funds to controlling shareholders.

6.3 | Alternative Channels

We next examine whether our main results could be explained by alternative mechanisms other than the proposed borrower-expectations channel. Detailed discussions and tests are reported in Section G of the Supporting Information. A first possibility is that PCA banks achieve lower defaults by restructuring loans more effectively, perhaps by credibly threatening costly insolvency proceedings. However, as discussed in Section 6.2.3, restructuring activity actually declines after banks enter PCA. Moreover, we test and find no significant difference in borrower performance after restructuring by PCA versus non-PCA banks, indicating that better restructuring outcomes are unlikely to explain the decline in defaults (Table G.1 in Section G).

A second explanation could be that PCA banks reduce defaults by posing a greater threat of referring delinquent borrowers to the IBC. We rule out this explanation by showing that reversals in defaults are no greater in jurisdictions with more efficient bankruptcy courts, where IBC threats would be more credible (Table G.2 in Section G).

A third possibility is that enhanced regulatory scrutiny under PCA improves banks' borrower screening and monitoring, mechanically leading to lower defaults. Two tests address this concern. First, Section 5.1 shows that default declines are unrelated to borrower health, as would be expected if banks were merely weeding out weaker clients. Second, we observe that defaults begin to reverse immediately after a PCA breach—even before formal admission into PCA—suggesting that borrower behavior responds to anticipated bank recovery rather than to operational improvements alone (Table G.3 in Section G).

7 | Conclusion

Extant literature has documented the existence of defaults in response to deteriorating bank health, particularly in environments with weak contract enforcement (Bond and Rai 2009; Schiantarelli et al. 2020). We show the existence of such defaults in the Indian setting. We then examine a regulatory intervention on banks by the Reserve Bank of India—PCA—to assess its impact on such defaults by borrowers. We find that PCA significantly reduces defaults on unhealthy banks. Our evidence suggests that this effect stems from its signaling value: borrowers expect the bank to recover, perceive greater benefit in maintaining the relationship, and consequently strengthen repayment discipline. We find that a special audit conducted by the central bank before PCA improved the timeliness of loan-loss provisions of banks. This improvement in reporting quality potentially enhanced the effectiveness of PCA by making the intervention more credible. Our study contributes to the literature on regulatory enforcement actions by shifting the focus beyond lender outcomes to borrower behavior, showing how such interventions influence repayment decisions.

Our study has certain limitations. A key limitation is that default is a rare event, and our results may therefore be influenced by a relatively small subset of observations. We also acknowledge that PCA entails short-term costs in the form of a substantial decline in lending. Our analysis focuses on the effect on defaults and does not attempt a full cost–benefit assessment. While PCA serves as a valuable early warning mechanism for identifying unhealthy banks and initiating corrective actions, it is not a one-size-fits-all solution. In cases of rapid deterioration arising from governance failures or bank runs, regulators may need to employ more urgent, ad hoc interventions. Moreover, our identification strategy rests on the assumption that borrowers monitor bank health and adjust their behavior accordingly—a mechanism consistent with our findings but not directly observable in our setting. Finally, while most default reversals appear consistent with borrower expectations of bank recovery, we cannot entirely dismiss the possibility that enhanced screening and monitoring, effective restructuring practices, or timely bankruptcy referrals also contributed to the observed decline in defaults.

Overall, our findings underscore the broad role of regulatory enforcement in sustaining financial stability by shaping borrower incentives, offering important insights for banking regulators and policymakers.

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Data Availability Statement

All data are available from the sources cited in the text.

Endnotes

- ¹ In the United States, enforcement actions against banks are typically carried out by the Federal Reserve. See the Federal Reserve's overview of enforcement actions on the Federal Reserve Bank website (<https://www.federalreserve.gov/supervisionreg/enforcement-actions-about.htm>), and a complete list of such actions on the enforcement actions website (<https://www.federalreserve.gov/supervisionreg/enforcementactions.htm>).
- ² Banking regulators worldwide use PCA, a type of enforcement action, to stabilize distressed banks and prevent their collapse (Jones and King 1995; Kleymentova and Tomy 2022; An et al. 2024).
- ³ Boehm and Oberfield (2020), Amirapu (2021), and Boehm (2022) document weaknesses in India's court system, underscored by the country's rank of 130 out of 189 in the World Bank's Ease of Doing Business contract enforcement index.
- ⁴ These costs are not necessarily mutually exclusive.
- ⁵ The "running variable" is the difference between a bank's actual value on the relevant metric and the corresponding PCA threshold. It determines treatment assignment based on whether the value is positive (treated) or negative (control).
- ⁶ The AQR is a rigorous special audit exercise conducted by the regulator to validate the actual level of non-performing loans and asset quality of the banks.
- ⁷ See Sapienza (2004); Correia (2014); Mukherjee et al. (2018); Koetter and Popov (2021); Knechel and Park (2022); Huang and Thakor (2024).
- ⁸ Loan evergreening refers to the practice of directly or indirectly re-lending to financially distressed borrowers to help them repay their existing loans and delay loan loss recognition.
- ⁹ See Chircop and Novotny-Farkas (2016); Ertan et al. (2017); Delis et al. (2017); Granja (2018); Corona et al. (2019); Costello et al. (2019); Wheeler (2019); Balakrishnan et al. (2021); Bertomeu et al. (2023); Mahieux et al. (2023); An et al. (2024); Granja and Leuz (2024).
- ¹⁰ The Indian financial year begins in April and ends in March. For example, FY2018 covers April 2017 to March 2018. Accordingly, the revised PCA, introduced on April 13, 2017, became effective from FY2018 onward.
- ¹¹ For instance, in FY2007, despite IDBI Bank and Dena Bank exceeding the NNPA threshold by ~30% and 10%, respectively, and the United Bank of India and the Central Bank of India surpassing the ROA limit by nearly 400%, none were placed under PCA.
- ¹² A Credit Suisse report finds that five banks breaching the Level I threshold in FY2018 were yet to be admitted under PCA (Mudgill 2018).
- ¹³ Please see the [Supporting Information](#).
- ¹⁴ Level III is a subset of Level II breaches, with banks violating them on only eight occasions. Hence, we treat Level II breaches as the operative trigger for PCA admission.
- ¹⁵ India's low ranking in contract enforcement persists despite major debt-recovery and bankruptcy laws such as the *Securitization and Reconstruction of Financial Assets and Enforcement of Security Interest Act* (SARFAESI) of 2002 (Vig 2013) and the IBC introduced in 2016 (Chakraborty et al. 2022; Hope et al. 2025).
- ¹⁶ India's GDP grew by an average of 6.7% in the 5 years preceding the onset of the COVID-19 pandemic.
- ¹⁷ We subsequently discuss the relevance of each of these economic forces in detail in Section 4.1.3.
- ¹⁸ Loans to agriculture and other specified sectors are considered priority sector loans.
- ¹⁹ In ~67% of firm-bank-quarter observations, firms have relationships with multiple banks.
- ²⁰ Of the 12 banks that breach PCA thresholds, seven are admitted in the same quarter, while five enter one to two quarters later—likely reflecting the time taken by the central bank to assess banks' financials. Table A.1 reports the breach and admission quarters.
- ²¹ The ROA cut-off is zero, and PCA is triggered when a bank reports a negative ROA for 3 consecutive years. We use the minimum ROA over the last 3 years to calculate the standardized ROA score.
- ²² We choose the default options prescribed in the manual by Calonico et al. (2017): bandwidth selection = "mserd" and vce estimator = "nn3." We find that our inferences are robust to all other methodologies.
- ²³ We also estimate the specification separately for subsamples with low and high values of firm health. As shown in Table A.3, the reversal in defaults remains unrelated to firm health.
- ²⁴ We conduct the above test at the portfolio level, not the stock level, because it allows us to group admitted banks based on media coverage on the event dates and examine how markets react to PCA announcements across these groups.

- ²⁵ Although this test directly examines changes in borrower beliefs, the limited number of bank observations constrains our ability to make strong statistical inferences. We therefore view these findings as suggestive and complement them with several additional tests of the underlying mechanism.
- ²⁶ We start our sample in FY2013 to ensure a sufficient number of pre-AQR periods for estimating a DiD regression to assess the effect of the AQR.
- ²⁷ We start our sample in 2016 so that we have 2 years of pre-PCA data for the DiD analysis.
- ²⁸ We test and rule out any pre-existing trends in Columns 1 and 2 of Table F.2.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.